

Introduction to SQL Project

NEW WHEELS DATA
ANNAPOORNA S

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New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

New Wheels: Data Dictionary

New Wheels Data Dictionary

shipper_id: Unique ID of the Shipper

shipper_name: Name of the Shipper

shipper_contact_details: Contact detail of the Shipper

product_id: Unique ID of the Product

vehicle_maker: Vehicle Manufacturing company name

vehicle_model: Vehicle model name

vehicle_color: Color of the Vehicle

vehicle_model_year: Year of Manufacturing

vehicle_price: Price of the Vehicle

quantity: Ordered Quantity

customer_id: Unique ID of the customer

customer_name: Name of the customer

gender: Gender of the customer

job_title: Job Title of the customer

phone_number: Contact detail of the customer

email_address: Email address of the customer

city: Residing city of the customer

country: Residing country of the customer

state: Residing state of the customer

customer_address: Address of the customer

order_date: Date on which customer ordered the vehicle

order_id: Unique ID of the order

ship_date: Shipment Date

ship_mode: Shipping Mode/Class

shipping: Shipping Ways

postal_code: Postal Code of the customer

discount: Discount given to the customer for the particular order by credit card in percentage

credit_card_type: Credit Card Type

credit_card_number: Credit card number

customer_feedback: Feedback of the customer

quarter_number : Quarter Number

New Wheels: ER Diagram

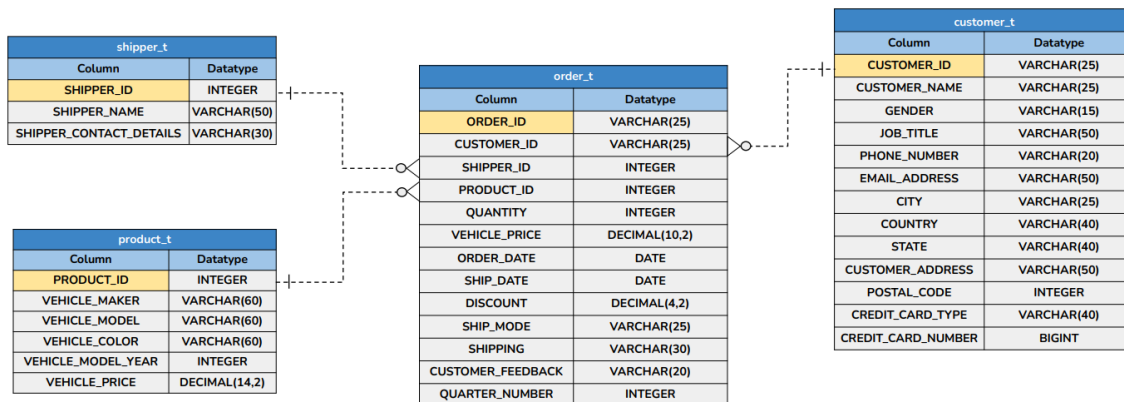


Figure 1: Entity-Relationship Diagram

Business Questions

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

a) `<SELECT count(CUSTOMER_ID) from customer_t;>`

b) `<`

```
SELECT c.state, COUNT(DISTINCT o.customer_id) AS total_customers
FROM customer_t c
```

```
JOIN order_t o ON c.customer_id = o.customer_id

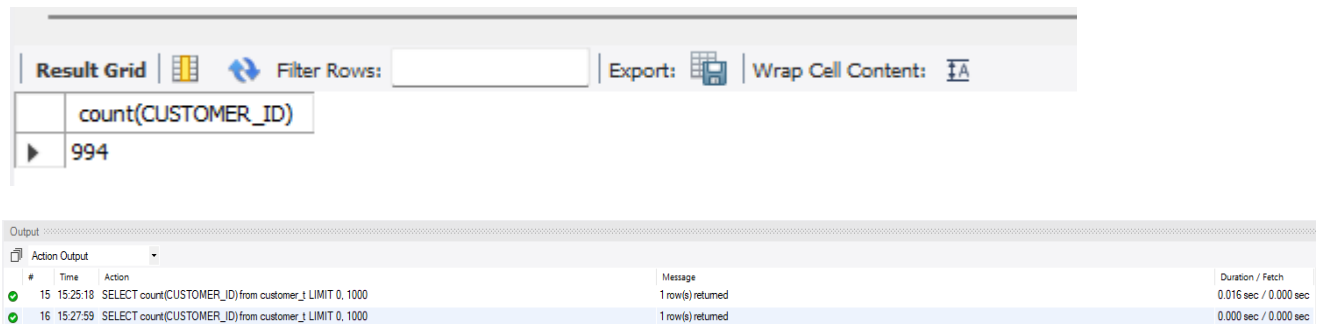
GROUP BY c.state

ORDER BY total_customers DESC;

>
```

Output:

a)



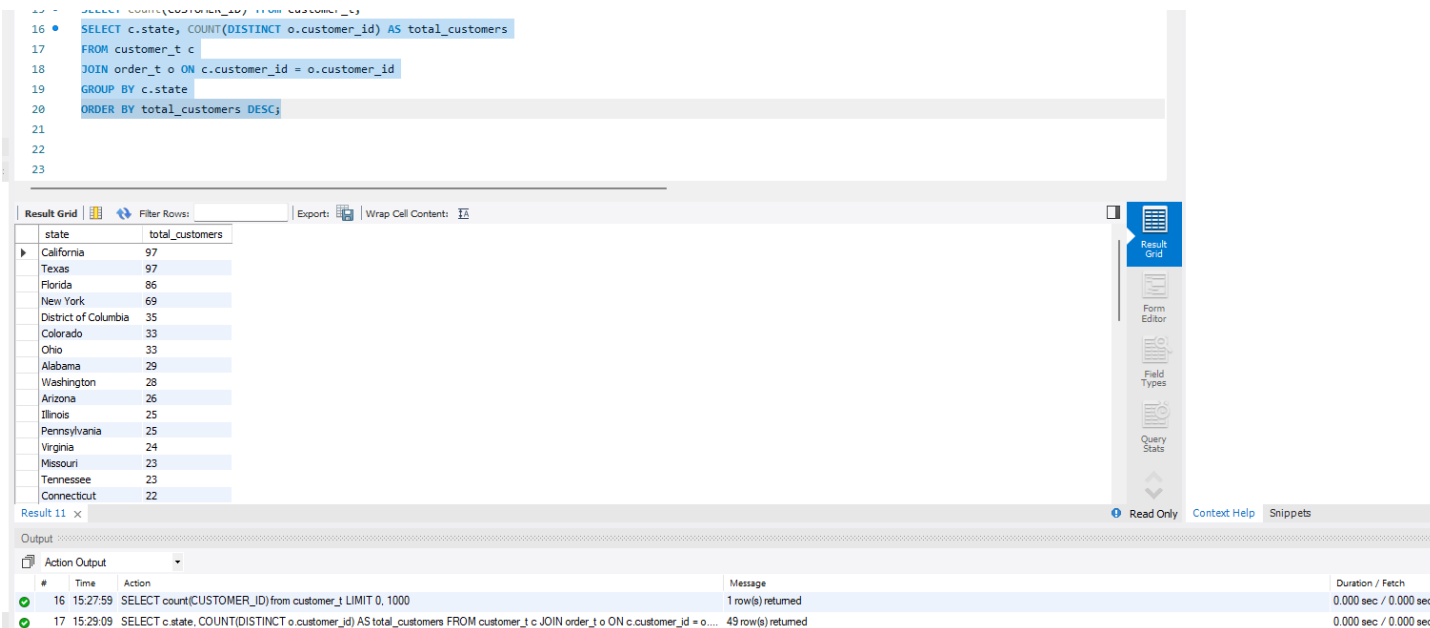
Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

count(CUSTOMER_ID)	
	994

Output

#	Time	Action	Message	Duration / Fetch
15	15:25:18	SELECT count(CUSTOMER_ID) from customer_t LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
16	15:27:59	SELECT count(CUSTOMER_ID) from customer_t LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

b)



Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

state	total_customers
California	97
Texas	97
Florida	86
New York	69
District of Columbia	35
Colorado	33
Ohio	33
Alabama	29
Washington	28
Arizona	26
Illinois	25
Pennsylvania	25
Virginia	24
Missouri	23
Tennessee	23
Connecticut	22

Result 11 x

Output

#	Time	Action	Message	Duration / Fetch
16	15:27:59	SELECT count(CUSTOMER_ID) from customer_t LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
17	15:29:09	SELECT c.state, COUNT(DISTINCT o.customer_id) AS total_customers FROM customer_t c JOIN order_t o ON c.customer_id = o.customer_id GROUP BY c.state ORDER BY total_customers DESC;	49 row(s) returned	0.000 sec / 0.000 sec

Figure 2: Output of Query1

Observations and Insights:

- Geographic Concentration Risk:** Out of 994 total customers, the top 3 states (California, Texas, Florida) account for 280 customers (28.2% of total customer base), creating significant revenue dependency. The top 10 states contain 558 customers (56.1%), while the bottom 10 states combined have only 46 customers (4.6%). This extreme concentration means business performance is heavily influenced by market conditions in just a few states.

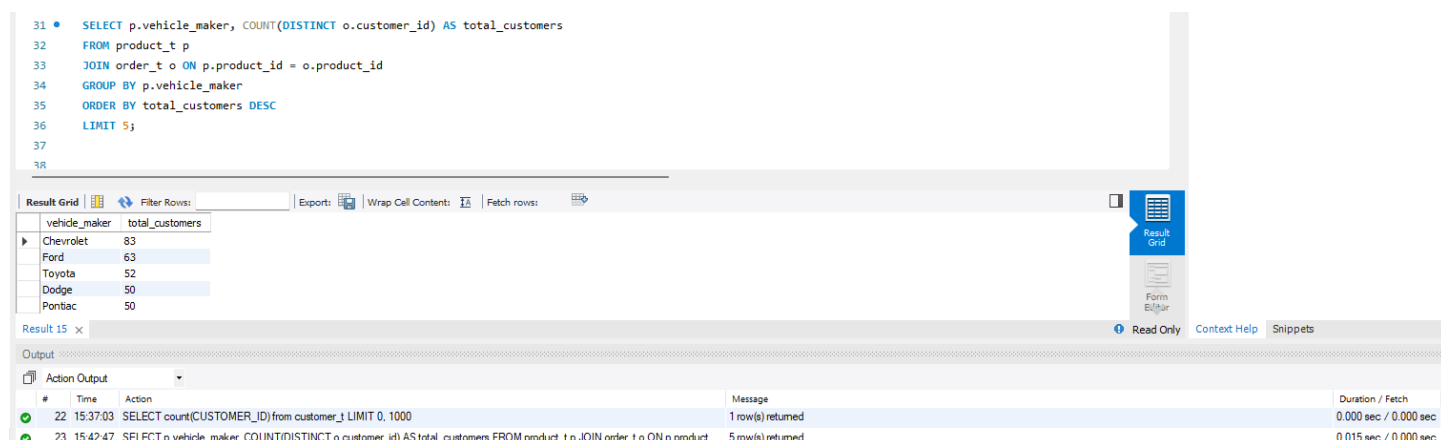
- **Uneven Market Penetration with Untapped Potential:** Stark disparity exists between highest-performing states (California/Texas at 97 customers each) and lowest-performing states (Maine, Vermont, Wyoming at 1 customer each)—a 97x difference. Notably, densely populated states like New Jersey (9 customers, 0.9%), Massachusetts (14 customers, 1.4%), and Wisconsin (8 customers, 0.8%) are significantly underperforming relative to their population size, indicating substantial growth opportunities. With 15 states having fewer than 10 customers, the business has considerable room for geographic expansion and diversification to reduce concentration risk.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
<SELECT p.vehicle_maker, COUNT(DISTINCT o.customer_id) AS total_customers
FROM product_t p
JOIN order_t o ON p.product_id = o.product_id
GROUP BY p.vehicle_maker
ORDER BY total_customers DESC
LIMIT 5;
>
```

Output:



vehicle_maker	total_customers
Chevrolet	83
Ford	63
Toyota	52
Dodge	50
Pontiac	50

Figure 3: Output of Query2

Observations and Insights:

- **Chevrolet Dominance with American Brand Preference:** Chevrolet leads significantly with 83 customers, representing a 32% advantage over second-place Ford (63 customers). American brands dominate the top 5, capturing 4 out of 5 positions (Chevrolet, Ford, Dodge, Pontiac) with 246 combined customers, while Toyota is the sole import brand with 52 customers. This indicates strong customer preference for domestic manufacturers, representing 82.6% of the top 5 customer base.

- **Moderate Brand Concentration with Growth Opportunity:** The top 5 vehicle makers collectively serve 298 unique customers out of 994 total (30% of customer base), suggesting relatively diversified brand preferences across the broader customer base. The gap between #1 Chevrolet (83) and #5 Pontiac (50) shows a 66% difference, indicating clear performance tiers. Toyota's strong showing as the only import demonstrates quality-conscious customers exist, presenting opportunity to expand import offerings and capture customers seeking alternatives to domestic brands.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
<WITH StateMakerCounts AS (  
    SELECT  
        c.state,  
        p.vehicle_maker,  
        COUNT(DISTINCT o.customer_id) AS customer_count,  
        RANK() OVER (PARTITION BY c.state ORDER BY COUNT(DISTINCT  
o.customer_id) DESC) AS maker_rank  
    FROM customer_t c  
    JOIN order_t o ON c.customer_id = o.customer_id  
    JOIN product_t p ON o.product_id = p.product_id  
    GROUP BY c.state, p.vehicle_maker  
)  
SELECT  
    state,  
    vehicle_maker,  
    customer_count,  
    maker_rank
```

```
FROM StateMakerCounts

WHERE maker_rank = 1

ORDER BY customer_count DESC;

>
```

Output:

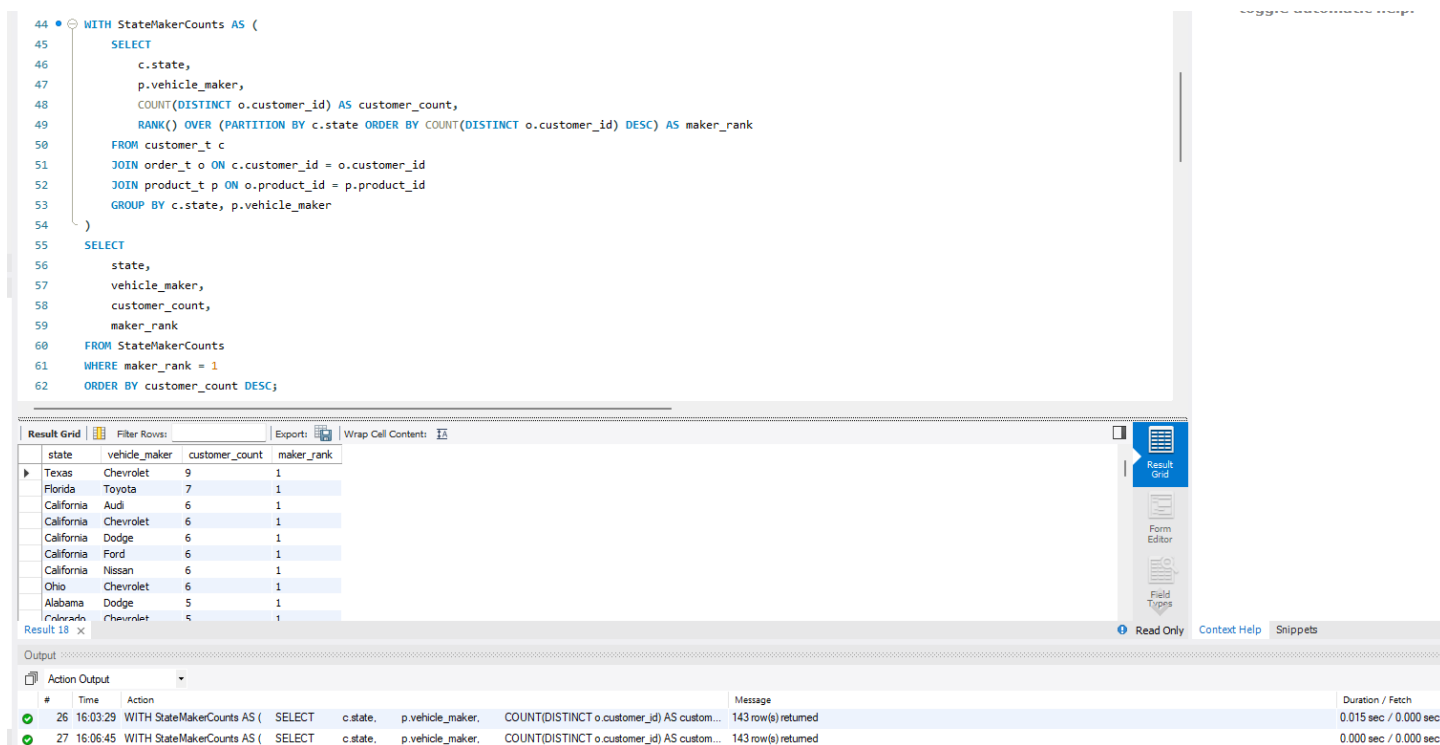


Figure 4: Output of Query3

Observations and Insights:

- Chevrolet's Geographic Dominance with Significant State Fragmentation:** Chevrolet appears as the sole or tied leader in 12 states (Texas, Ohio, Colorado, Washington, District of Columbia, Missouri, Illinois, Alaska, Connecticut, Massachusetts, and others), confirming its national dominance from Question 2. However, 22 states (45% of all markets) exhibit multi-way ties for rank 1, with extreme cases like California (5-way tie), Iowa (11-way tie), and Kansas (13-way tie). This fragmentation indicates that while Chevrolet leads nationally, nearly half of state markets lack clear brand preference due to either competitive saturation or insufficient market penetration with low customer counts (1-2 customers per maker).
- Clear Regional Brand Preferences: Domestic Heartland vs. Import Coastal Markets:** Distinct geographic patterns emerge—inland/heartland states strongly prefer domestic brands (Texas: Chevrolet-9, Ohio: Chevrolet-6, Colorado: Chevrolet-5, Alabama: Dodge-5), while coastal/urban markets favor imports or show diversity (Florida: Toyota-7, New York: Toyota/Pontiac-5, Georgia:



Toyota-3, Pennsylvania: Toyota-3). California's 5-way tie including imports (Audi, Nissan) alongside domestics reflects its diverse, competitive market. Small states with extreme fragmentation (Kansas 13-way, Iowa 11-way, Wisconsin 8-way all at 1 customer each) signal underdeveloped markets requiring targeted penetration strategies rather than broad inventory approaches.

Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
<SELECT
    'Overall' AS quarter_number,
    ROUND (AVG (
        CASE
            WHEN customer_feedback = 'Very Bad' THEN 1
            WHEN customer_feedback = 'Bad' THEN 2
            WHEN customer_feedback = 'Okay' THEN 3
            WHEN customer_feedback = 'Good' THEN 4
            WHEN customer_feedback = 'Very Good' THEN 5
        END
    ), 2) AS avg_rating
FROM order_t
WHERE customer_feedback IS NOT NULL

UNION ALL

SELECT
    CAST(quarter_number AS CHAR) AS quarter_number,
```



```
ROUND (AVG (
```

```
    CASE
```

```
        WHEN customer_feedback = 'Very Bad' THEN 1
```

```
        WHEN customer_feedback = 'Bad' THEN 2
```

```
        WHEN customer_feedback = 'Okay' THEN 3
```

```
        WHEN customer_feedback = 'Good' THEN 4
```

```
        WHEN customer_feedback = 'Very Good' THEN 5
```

```
    END
```

```
), 2) AS avg_rating
```

```
FROM order_t
```

```
WHERE customer_feedback IS NOT NULL
```

```
GROUP BY quarter_number
```

```
ORDER BY quarter_number;
```

```
>
```

Output:

```

102 SELECT
103     'Overall' AS quarter_number,
104     ROUND(AVG(
105         CASE
106             WHEN customer_feedback = 'Very Bad' THEN 1
107             WHEN customer_feedback = 'Bad' THEN 2
108             WHEN customer_feedback = 'Okay' THEN 3
109             WHEN customer_feedback = 'Good' THEN 4
110             WHEN customer_feedback = 'Very Good' THEN 5
111         END
112     ), 2) AS avg_rating
113 FROM order_t
114 WHERE customer_feedback IS NOT NULL
115
116 UNION ALL
117
118 SELECT
119     CAST(quarter_number AS CHAR) AS quarter_number,
120     ROUND(AVG(
121         CASE
122             WHEN customer_feedback = 'Very Bad' THEN 1
123             WHEN customer_feedback = 'Bad' THEN 2
124             WHEN customer_feedback = 'Okay' THEN 3
125             WHEN customer_feedback = 'Good' THEN 4
126             WHEN customer_feedback = 'Very Good' THEN 5
127         END
128     ), 2) AS avg_rating
129 FROM order_t
130 WHERE customer_feedback IS NOT NULL
131 GROUP BY quarter_number
132 ORDER BY quarter_number;
133

```

quarter_number	avg_rating
1	3.55
2	3.35
3	2.96
4	2.40
Overall	3.14

Result 24 x

Output

#	Time	Action	Message
32	16:14:55	SELECT quarter_number, ROUND(AVG(CASE WHEN customer_feedback = 'Very Bad' THEN 1 WHEN...	4 row(s) returned
33	16:15:03	SELECT 'Overall' AS quarter_number, ROUND(AVG(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ...	5 row(s) returned

Figure 5: Output of Query4

Observations and Insights:

- **Critical Declining Customer Satisfaction Trend:** Customer ratings show a steep and consistent decline across all four quarters, dropping from 3.55 in Q1 to 2.40 in Q4—a 32% decrease representing 1.15 rating points lost. The overall average of 3.14 (between "Okay" and "Good") masks this alarming deterioration. Each quarter shows sequential decline (Q1→Q2: -0.20, Q2→Q3: -0.39, Q3→Q4: -0.56), with the steepest drop occurring in Q4. By Q4, the average rating of 2.40 falls below "Okay" (3.0) and approaches "Bad" (2.0), indicating severe customer dissatisfaction and potential business crisis requiring immediate intervention.
- **Root Cause Investigation and Business Impact Urgency:** The consistent quarterly degradation (no recovery in any quarter) suggests systemic operational issues rather than isolated incidents—potential causes include deteriorating product quality, declining service levels, shipping delays, inventory/stock

issues, or poor post-sale support. With Q4 rating at 2.40, the business faces high churn risk, negative word-of-mouth, and revenue loss. The 32% satisfaction drop correlates with potential customer retention problems that could manifest in reduced repeat purchases and damaged brand reputation. Immediate action is required: conduct customer surveys, review Q3-Q4 operational changes, analyze negative feedback themes, and implement corrective measures to reverse this trend before Q5.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
<SELECT
    quarter_number,
    COUNT(*) AS total_feedback,
    ROUND(SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) *
100.0 / COUNT(*), 2) AS very_bad_pct,
    ROUND(SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) * 100.0
/ COUNT(*), 2) AS bad_pct,
    ROUND(SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) *
100.0 / COUNT(*), 2) AS okay_pct,
    ROUND(SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) *
100.0 / COUNT(*), 2) AS good_pct,
    ROUND(SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) *
100.0 / COUNT(*), 2) AS very_good_pct,
    ROUND((SUM(CASE WHEN customer_feedback IN ('Bad', 'Very Bad') THEN 1
ELSE 0 END) * 100.0 / COUNT(*)), 2) AS dissatisfaction_rate,
    ROUND((SUM(CASE WHEN customer_feedback IN ('Good', 'Very Good') THEN 1
ELSE 0 END) * 100.0 / COUNT(*)), 2) AS satisfaction_rate
FROM order_t
WHERE customer_feedback IS NOT NULL
GROUP BY quarter_number
ORDER BY quarter_number;
>
```

Output:

```

141 • SELECT
142     quarter_number,
143     COUNT(*) AS total_feedback,
144     ROUND(SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS very_bad_pct,
145     ROUND(SUM(CASE WHEN customer_feedback = 'Bad' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS bad_pct,
146     ROUND(SUM(CASE WHEN customer_feedback = 'Okay' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS okay_pct,
147     ROUND(SUM(CASE WHEN customer_feedback = 'Good' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS good_pct,
148     ROUND(SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2) AS very_good_pct,
149     ROUND((SUM(CASE WHEN customer_feedback IN ('Bad', 'Very Bad') THEN 1 ELSE 0 END) * 100.0 / COUNT(*)), 2) AS dissatisfaction_rate,
150     ROUND((SUM(CASE WHEN customer_feedback IN ('Good', 'Very Good') THEN 1 ELSE 0 END) * 100.0 / COUNT(*)), 2) AS satisfaction_rate
151 FROM order_t
152 WHERE customer_feedback IS NOT NULL
153 GROUP BY quarter_number
154 ORDER BY quarter_number;

```

	quarter_number	total_feedback	very_bad_pct	bad_pct	okay_pct	good_pct	very_good_pct	dissatisfaction_rate	satisfaction_rate
1	310	10.97	11.29	19.03	28.71	30.00	22.26	58.71	
2	262	14.89	14.12	20.23	22.14	28.63	29.01	50.76	
3	229	17.90	22.71	21.83	20.96	16.59	40.61	37.55	
4	199	30.65	29.15	20.10	10.05	10.05	59.80	20.10	

Result 25 x

#	Time	Action	Message
33	16:15:03	SELECT 'Overall' AS quarter_number, ROUND(AVG(CASE WHEN customer_feedback = 'Very Bad' THEN 1 ...	5 row(s) returned
34	16:19:26	SELECT quarter_number, COUNT(*) AS total_feedback, ROUND(SUM(CASE WHEN customer_feedback = 'Very Bad' TH...	4 row(s) returned

Figure 6: Output of Query5

Observations and Insights:

- YES - Severe and Accelerating Customer Dissatisfaction Crisis:** Dissatisfaction rate (Bad + Very Bad) has nearly tripled from Q1 (22.26%) to Q4 (59.80%), representing a 169% increase. Simultaneously, satisfaction rate (Good + Very Good) has collapsed from 58.71% in Q1 to just 20.10% in Q4—a 66% decline. Q4's dissatisfaction of 59.80% means nearly 6 out of 10 customers are actively unhappy. The "Very Bad" category alone exploded from 10.97% (Q1) to 30.65% (Q4), indicating extreme customer anger. The trend shows consistent quarterly deterioration with no signs of stabilization, and declining total feedback counts (310→199) suggest customers may be abandoning the business entirely rather than leaving feedback.
- Systematic Service/Quality Failure with Urgent Intervention Required:** The dramatic shift from positive to negative sentiment reveals fundamental operational breakdown rather than minor issues. "Very Good" ratings plummeted from 30% (Q1) to 10.05% (Q4), while "Very Bad" ratings surged 179% in the same period. The neutral "Okay" category remained stable (~20%), indicating problems aren't subjective—customers clearly experienced deteriorating service/product quality. With Q4 showing 3:1 dissatisfaction-to-satisfaction ratio (59.80% vs 20.10%), the business faces imminent reputational damage, customer churn, and revenue collapse. Root causes must be identified

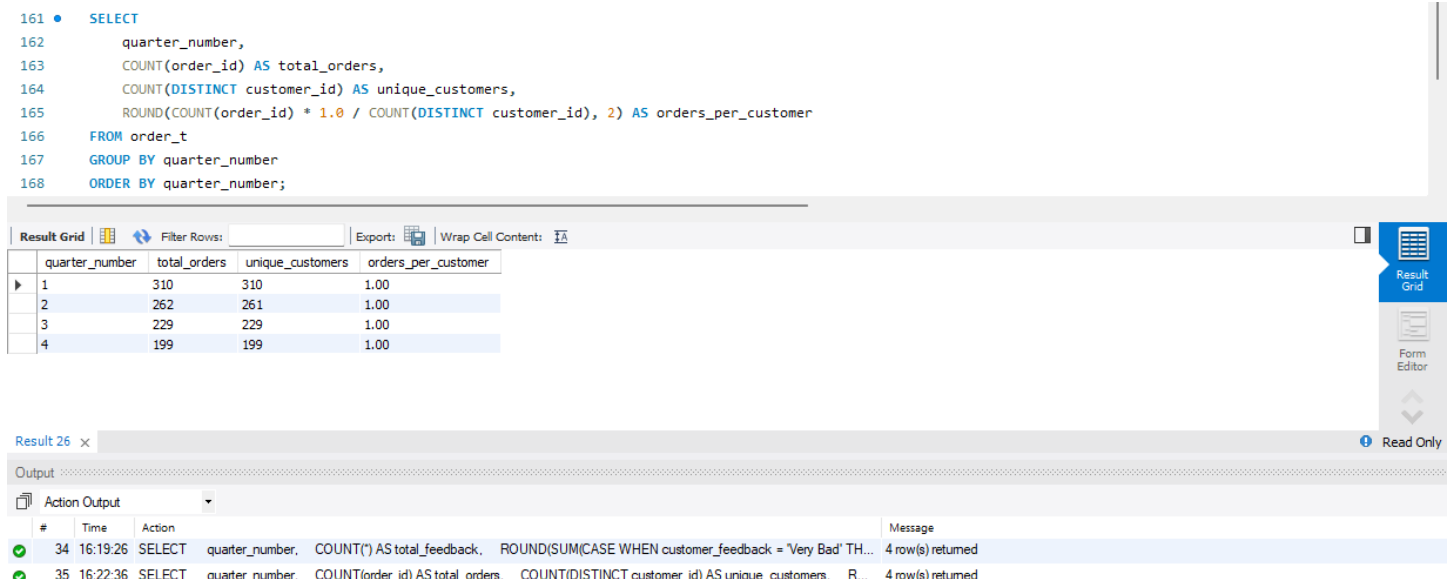
immediately: analyze Q2-Q3 operational changes (suppliers, shipping partners, staffing cuts, inventory quality), conduct exit surveys with dissatisfied customers, and implement emergency service recovery protocols to prevent complete customer base erosion.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
<SELECT
    quarter_number,
    COUNT(order_id) AS total_orders,
    COUNT(DISTINCT customer_id) AS unique_customers,
    ROUND(COUNT(order_id) * 1.0 / COUNT(DISTINCT customer_id), 2) AS
orders_per_customer
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
>
```

Output:



The screenshot displays a SQL query execution environment. The query is as follows:

```
161 • SELECT
162     quarter_number,
163     COUNT(order_id) AS total_orders,
164     COUNT(DISTINCT customer_id) AS unique_customers,
165     ROUND(COUNT(order_id) * 1.0 / COUNT(DISTINCT customer_id), 2) AS orders_per_customer
166 FROM order_t
167 GROUP BY quarter_number
168 ORDER BY quarter_number;
```

The results are shown in a table with the following data:

quarter_number	total_orders	unique_customers	orders_per_customer
1	310	310	1.00
2	262	261	1.00
3	229	229	1.00
4	199	199	1.00

Below the table, the 'Output' section shows the execution log with two entries:

#	Time	Action	Message
34	16:19:26	SELECT quarter_number, COUNT(*) AS total_feedback, ROUND(SUM(CASE WHEN customer_feedback = 'Very Bad' TH...	4 row(s) returned
35	16:22:36	SELECT quarter_number, COUNT(order_id) AS total_orders, COUNT(DISTINCT customer_id) AS unique_customers, R...	4 row(s) returned

Figure 7: Output of Query6

Observations and Insights:

- **Declining Order Volume with 36% Drop - Business Contraction Underway:** Order volume shows consistent quarterly decline from 310 orders (Q1) to 199 orders (Q4), representing a 36% decrease

(111 fewer orders). Each quarter experiences sequential decline: Q1→Q2 (-15.5%), Q2→Q3 (-12.6%), Q3→Q4 (-13.1%). This downward trend directly correlates with declining customer satisfaction observed in Questions 4 and 5, confirming that poor service/product quality is driving customers away. The business is losing approximately 37 orders per quarter on average, projecting potential Q5 volume below 170 orders if trend continues unchecked.

- **Zero Customer Loyalty - Critical Repeat Business Failure:** Orders per customer ratio of 1.00 across all quarters reveals a catastrophic finding: virtually no customers are making repeat purchases. With total orders nearly equal to unique customers each quarter (Q2: 262 orders from 261 customers shows only 1 repeat customer), New Wheels operates as a one-time transaction business with zero loyalty. This explains the declining order volume—the business must constantly acquire new customers to replace churning ones, and acquisition is failing to keep pace with attrition. Combined with 59.80% dissatisfaction rate (Q4, Question 5), dissatisfied customers aren't returning and aren't being replaced. Urgent need to implement retention strategies, loyalty programs, post-purchase follow-up, and service quality improvements to convert one-time buyers into repeat customers and reverse the declining order trajectory.

Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
<WITH QuarterlyRevenue AS (

    SELECT

        quarter_number,

        ROUND(SUM(quantity * (vehicle_price - ((discount / 100) *
vehicle_price))), 2) AS net_revenue,

        COUNT(order_id) AS total_orders

    FROM order_t

    GROUP BY quarter_number

)

SELECT

    quarter_number,

    net_revenue,

    total_orders,

    LAG(net_revenue) OVER (ORDER BY quarter_number) AS
previous_quarter_revenue,
```

```

ROUND (

    ((net_revenue - LAG(net_revenue) OVER (ORDER BY
quarter_number))) /

    LAG(net_revenue) OVER (ORDER BY quarter_number)) * 100, 2

) AS qoq_revenue_change_pct

FROM QuarterlyRevenue

ORDER BY quarter_number;

>

```

Output:

```

180 • WITH QuarterlyRevenue AS (
181     SELECT
182         quarter_number,
183         ROUND(SUM(quantity * (vehicle_price - ((discount / 100) * vehicle_price))), 2) AS net_revenue,
184         COUNT(order_id) AS total_orders
185     FROM order_t
186     GROUP BY quarter_number
187 )
188 SELECT
189     quarter_number,
190     net_revenue,
191     total_orders,
192     LAG(net_revenue) OVER (ORDER BY quarter_number) AS previous_quarter_revenue,
193     ROUND(
194         ((net_revenue - LAG(net_revenue) OVER (ORDER BY quarter_number))) /
195         LAG(net_revenue) OVER (ORDER BY quarter_number)) * 100, 2
196     ) AS qoq_revenue_change_pct
197 FROM QuarterlyRevenue
198 ORDER BY quarter_number;

```

Result Grid Filter Rows: Export: Wrap Cell Content:					
	quarter_number	net_revenue	total_orders	previous_quarter_revenue	qoq_revenue_change_pct
▶	1	39421580.16	310	NULL	NULL
	2	32715830.34	262	39421580.16	-17.01
	3	29229896.19	229	32715830.34	-10.66
	4	23346779.63	199	29229896.19	-20.13

Result 27 ×

Output

Action Output

#	Time	Action	Message
✓ 35	16:22:36	SELECT quarter_number, COUNT(order_id) AS total_orders, COUNT(DISTINCT customer_id) AS unique_customers, R...	4 row(s) returned
✓ 36	16:25:41	WITH QuarterlyRevenue AS (SELECT quarter_number, ROUND(SUM(quantity * (vehicle_price - ((discount / 100) * v...	4 row(s) returned

Figure 8: Output of Query7

Observations and Insights:

- **Revenue Collapse of 41% - Business in Severe Financial Decline:** Net revenue declined from \$39.4M (Q1) to \$23.3M (Q4), representing a devastating 41% drop (\$16.1M lost). Every quarter shows negative growth: Q2 (-17.01%), Q3 (-10.66%), Q4 (-20.13%), with Q4 experiencing the steepest decline. The business lost \$6.7M in Q2 alone, followed by additional losses of \$3.5M (Q3) and \$5.9M (Q4). This consistent quarterly deterioration—losing approximately \$5.4M per quarter on average—projects Q5 revenue could fall below \$19M if the trend continues, threatening business viability and potential insolvency.
- **Revenue Crisis Driven by Combined Volume and Satisfaction Collapse:** The 41% revenue decline directly correlates with the 36% order volume drop (Question 6) and 59.80% dissatisfaction rate (Question 5), confirming a vicious cycle: poor service → dissatisfied customers → no repeat purchases → declining orders → plummeting revenue. Average revenue per order (\$127,166 in Q1 vs \$117,322 in Q4) also declined 8%, suggesting customers are either buying cheaper vehicles or demanding larger discounts due to quality concerns. With zero customer loyalty (1.00 orders per customer, Question 6) and accelerating customer dissatisfaction, the business cannot sustain operations without immediate intervention. Critical actions needed: halt the service quality deterioration, implement customer win-back programs, review discount strategies to protect margins, and develop retention initiatives to stabilize revenue before Q5 results in catastrophic losses requiring workforce reductions or business closure.

Question 8: What is the trend of net revenue and orders by quarters?**Solution Query:**

```
< SELECT

    quarter_number,

    COUNT(order_id) AS total_orders,

    ROUND(SUM(quantity * (vehicle_price - ((discount / 100) *
vehicle_price))), 2) AS net_revenue,

    ROUND(AVG(quantity * (vehicle_price - ((discount / 100) *
vehicle_price))), 2) AS avg_order_value,

    ROUND(SUM(quantity), 0) AS total_units_sold

FROM order_t

GROUP BY quarter_number

ORDER BY quarter_number;

>
```

Output:

```

205 • SELECT
206     quarter_number,
207     COUNT(order_id) AS total_orders,
208     ROUND(SUM(quantity * (vehicle_price - ((discount / 100) * vehicle_price))), 2) AS net_revenue,
209     ROUND(AVG(quantity * (vehicle_price - ((discount / 100) * vehicle_price))), 2) AS avg_order_value,
210     ROUND(SUM(quantity), 0) AS total_units_sold
211 FROM order_t
212 GROUP BY quarter_number
213 ORDER BY quarter_number;

```

Result Grid					
Filter Rows:					
Export: Wrap Cell Content:					
	quarter_number	total_orders	net_revenue	avg_order_value	total_units_sold
▶	1	310	39421580.16	127166.39	464
	2	262	32715830.34	124869.58	400
	3	229	29229896.19	127641.47	342
	4	199	23346779.63	117320.50	306

Result 28 ×				
Output				
Action Output				
#	Time	Action	Message	
✓ 36	16:25:41	WITH QuarterlyRevenue AS (SELECT quarter_number, ROUND(SUM(quantity * (vehicle_price - ((discount / 100) * v...	4 row(s) returned	
✓ 37	16:28:59	SELECT quarter_number, COUNT(order_id) AS total_orders, ROUND(SUM(quantity * (vehicle_price - ((discount / 100) * ve...	4 row(s) returned	

Figure 9: Output of Query8

Observations and Insights:

- Parallel Decline in All Key Metrics - Comprehensive Business Deterioration:** Both revenue and orders show synchronized quarterly decline: orders dropped 36% (310→199) while revenue fell 41% (\$39.4M→\$23.3M), confirming strong correlation between volume and revenue performance. Total units sold declined 34% (464→306 vehicles), matching order decline and indicating minimal multi-vehicle purchases. The steeper revenue decline (41%) versus order decline (36%) reveals margin erosion, with average order value dropping 7.7% from \$127,166 (Q1) to \$117,321 (Q4). Q3 showed temporary avg order value recovery to \$127,641, but Q4's sharp drop to \$117,321 signals accelerating problems with customers shifting to lower-priced vehicles or demanding higher discounts.
- Business Model Breakdown - Revenue Decline Outpacing Volume Loss Indicates Pricing/Margin Crisis:** The fact that revenue declined 5% more than order volume (41% vs 36%) demonstrates pricing power erosion—customers are either negotiating better deals or selecting cheaper inventory due to budget constraints or dissatisfaction. With avg order value falling \$9,846 from Q1 to Q4, each lost order compounds into disproportionate revenue impact. The business is trapped in a death spiral: dissatisfaction (59.80%, Q5) → lost customers → declining orders (Q6) → revenue collapse (Q7/Q8)

→ potential cost-cutting that further degrades service → more dissatisfaction.

Immediate strategic pivot required: stop the discount spiral, focus on high-margin premium vehicles, implement service recovery to protect remaining customer base, and shift from volume-chasing to value-preservation to stabilize per-order economics before addressing growth.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
< SELECT
    c.credit_card_type,
    ROUND(AVG(o.discount), 2) AS avg_discount_pct
FROM order_t o
JOIN customer_t c ON o.customer_id = c.customer_id
GROUP BY 1
ORDER BY avg_discount_pct DESC;
>
```

Output:

```

221 • SELECT
222     c.credit_card_type,
223     ROUND(AVG(o.discount), 2) AS avg_discount_pct
224 FROM order_t o
225 JOIN customer_t c ON o.customer_id = c.customer_id
226 GROUP BY 1
227 ORDER BY avg_discount_pct DESC;

```

credit_card_type	avg_discount_pct
laser	0.64
mastercard	0.63
visa-electron	0.62
china-unionpay	0.62
americanexpress	0.62
maestro	0.62
instapayment	0.62
jcb	0.61

Result 29 x

Output

Action Output

#	Time	Action	Message
37	16:28:59	SELECT quarter_number, COUNT(order_id) AS total_orders, ROUND(SUM(quantity * (vehicle_price - ((discount / 100) * ve...	4 row(s) returned
38	16:35:42	SELECT c.credit_card_type, ROUND(AVG(o.discount), 2) AS avg_discount_pct FROM order_t o JOIN customer_t c ON o.cu...	16 row(s) returned

Figure 10: Output of Query9

Observations and Insights:

• Extremely Low and Uniform Discount Structure - Minimal Payment Type

Differentiation: Average discounts across all 16 credit card types range from 0.58% to 0.64%, showing only 0.06 percentage points variation—a mere 10% difference between highest and lowest. These extremely low discount rates (all under 1%) indicate discounts are NOT tied to payment method but rather applied uniformly across transactions regardless of credit card type. The tight clustering (0.60-0.62% for most cards) suggests random distribution rather than strategic payment partnerships, meaning New Wheels isn't leveraging payment processors for promotional partnerships or incentive programs.

• Missed Revenue Opportunity and Strategic Implications: The lack of differentiation reveals New Wheels isn't using payment-based discount strategies to influence customer behavior (e.g., encouraging lower-fee payment methods, partnering with specific card networks for co-marketing). With average discounts below 1%, payment method discounts contribute minimally to the overall business discount strategy or revenue erosion observed in Questions 7-8. The uniform distribution indicates either: (1) discounts are applied at product/customer level rather than payment level, or (2) system applies small transaction fees rather than promotional discounts. Recommendation: explore strategic partnerships with major card networks (Visa, Mastercard, AmEx) for co-branded promotions,

cashback programs, or exclusive financing offers that could drive customer acquisition without eroding margins, while potentially reducing transaction processing fees through volume agreements.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
< SELECT
    quarter_number,
    ROUND(AVG(datediff(ship_date, order_date)), 2) AS avg_shipping_days,
    COUNT(order_id) AS total_orders
FROM order_t
WHERE ship_date IS NOT NULL AND order_date IS NOT NULL
GROUP BY quarter_number
ORDER BY quarter_number;
>
```

Output:

```

234 • SELECT
235     quarter_number,
236     ROUND(AVG(datediff(ship_date, order_date)), 2) AS avg_shipping_days,
237     COUNT(order_id) AS total_orders
238 FROM order_t
239 WHERE ship_date IS NOT NULL AND order_date IS NOT NULL
240 GROUP BY quarter_number
241 ORDER BY quarter_number;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	quarter_number	avg_shipping_days	total_orders
▶	1	57.17	310
	2	71.11	262
	3	117.76	229
	4	174.10	199

Result 32 x

Output

Action Output

#	Time	Action	Message
✓ 41	16:43:22	SELECT c.credit_card_type, ROUND(AVG(o.discount), 2) AS avg_discount_pct FROM order_t o JOIN customer_t c ON o.cu...	16 row(s) returned
✓ 42	16:43:28	SELECT quarter_number, ROUND(AVG(datediff(ship_date, order_date)), 2) AS avg_shipping_days, COUNT(order_id) AS t...	4 row(s) returned

Figure 11: Output of Query10

Observations and Insights:

- Catastrophic Shipping Delays - 204% Increase Destroying Customer Satisfaction:** Average shipping time exploded from 57 days (Q1) to 174 days (Q4), representing a 204% increase and adding 117 additional days to fulfillment. Customers now wait nearly 6 months (174 days) from order to delivery in Q4, compared to roughly 2 months in Q1. The deterioration accelerates each quarter: Q1→Q2 (+24%), Q2→Q3 (+66%), Q3→Q4 (+48%). This catastrophic shipping performance directly explains the customer satisfaction collapse observed in Questions 4-5: Q4's 59.80% dissatisfaction rate and 2.40 average rating are inevitable outcomes when customers wait half a year for delivery. Industry standard for vehicle delivery is typically 30-60 days; New Wheels is operating at 3-6x acceptable timeframes.
- Root Cause of Business Failure Identified - Operational Breakdown Driving All Negative Trends:** The shipping delay crisis is the smoking gun explaining the entire business deterioration chain: prolonged wait times (Q10) → extreme dissatisfaction (Q4-Q5) → zero repeat purchases (Q6) → declining orders (Q6) → revenue collapse (Q7-Q8). Q4's 174-day average means customers ordering in January receive vehicles in June/July, creating unacceptable service levels that guarantee negative reviews, refund requests, and customer churn. **CRITICAL PRIORITY: This is a business-**

ending emergency requiring immediate executive intervention. Actions needed: identify logistics/supply chain bottlenecks (inventory shortages, shipping partner failures, order processing delays), implement express fulfillment for new orders, proactively communicate with customers in shipping queue, offer compensation/upgrades for delayed orders, and establish 30-45 day shipping SLA target to salvage Q5 and prevent total business collapse.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
124,714,086.32	1000	994	3.14
Last Quarter Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
23,346,779.63	199	174.10	20.10%

Table:

Table 1: Business metrics Overview

Business Recommendations

- **URGENT: Fix Shipping Crisis Immediately** - Q4 shipping time of 174 days (nearly 6 months) is the root cause of all business failures. Conduct emergency audit of supply chain, logistics partners, and inventory processes. Implement express fulfillment protocols to reduce shipping to 30-45 days maximum. Proactively communicate with customers awaiting delivery and offer compensation for delays to prevent further reputation damage.
- **Launch Customer Win-Back and Retention Program** - With 59.80% dissatisfaction rate, zero repeat purchases (1.00 orders per customer), and 36% order decline, implement immediate service recovery: personalized outreach to dissatisfied Q3-Q4 customers, loyalty discounts for repeat purchases, post-delivery follow-up surveys, and quality assurance checkpoints. Target converting one-time buyers into repeat customers to stabilize declining order volume.
- **Strategic Pivot to Stabilize Revenue** - Revenue collapsed 41% from \$39.4M (Q1) to \$23.3M (Q4) due to combined volume loss and margin erosion. Stop aggressive discounting, focus inventory on high-margin vehicles (Chevrolet, Toyota, Ford based on customer preferences), strengthen partnerships with top manufacturers, and prioritize profitability over volume. Allocate marketing budget to high-performing states (CA, TX, FL) and protect core customer base before pursuing expansion.